



7. Programme Structure

Discipline Core Courses (C)									
Course Code	Course Title	Hours/Week				L	T	P	C
		L	T	P	C				
UCS24101J	Digital Electronics	3	0	3	4				
USA24102J	Programming for Problem Solving	3	0	3	4				
UMS24101T	Discrete Mathematical Structures	4	0	0	4				
USA24201J	Data Structures and Algorithms	3	0	3	4				
UCS24202J	Object Oriented Programming using Java	3	0	3	4				
UMS24202T	Mathematical Foundation	4	0	0	4				
UCS24301J	Operating System	3	0	3	4				
USA24302J	Database Management System	3	0	3	4				
UMS24303T	Statistical Methods	4	0	0	4				
USA24401J	Python Programming	3	0	3	4				
Total Learning Credits									40

Discipline Core/ Elective Courses (D)									
Course Code	Course Title	Hours/Week				L	T	P	C
		L	T	P	C				
UMS24D01T	Resource Management Techniques	4	0	0	4				
UMS24D02T	Numerical Methods								
UCS24D01J	Software Engineering								
UCS24D02J	Computer Networks and Data Communication	3	0	3	4				
UCS24D03J	Object Oriented Analysis and Design	3	0	3	4				
UCS24D04J	Machine Learning								
UCS24D05J	Fundamentals of Business Application Development	3	0	3	4				
UCS24D06J	NLP and Language Models								
USA24D07J	Cloud Computing	3	0	3	4				
UCS24D08J	Microprocessors and Embedded Systems								
UCS24D09J	Mobile Application Development	3	0	3	4				
USA24D10T	Wireless Communication and Mobile Computing	4	0	0	4				
UCS24D11J	Web Application Framework								
UCS24D12J	Computer Vision	3	0	2	4				
UCS24D13J	Big Data Analytics								
UCS24D14J	Cryptography and Network Security	3	0	2	4				
UCS24D15J	AI and Business Analytics								
USA24D18J	Cyber Security	3	0	2	4				
USA24D20J	Blockchain Technology	3	0	2	4				
UCS24D18T	Introduction to Cyber Physical Systems	4	0	0	4				
Total Learning Credits									40

Elective Courses (Minor) (E)									
Course Code	Course Title	Hours/Week				L	T	P	C
		L	T	P	C				
UDS24E01J	Data Science and Analytics	3	0	3	4				
UDS24E02J	Data Analytics using Spreadsheet	3	0	3	4				
UDS24E03J	Essentials of Machine Learning	3	0	3	4				
UDS24E04J	Programming using R	3	0	3	4				
UDS24E05J	Programming using Python	3	0	3	4				
UDS24E06J	Introduction to Deep Learning	3	0	3	4				
UDS24E07J	Applications of Computer Vision	3	0	3	4				
UDS24E08J	Data Visualization Tools	3	0	3	4				
Total Learning Credits									32

Multidisciplinary Courses (M)									
Course Code	Course Title	Hours/Week				L	T	P	C
		L	T	P	C				
UCS24M01J	Introduction to ICT Tools	2	0	2	3				
UCS24M02J	Fundamentals of Digital Marketing	3	0	1	3				
UCS24M03J	Basics of Web Application Development	2	0	2	3				
Total Learning Credits									9

Value Added Courses (V)									
Course Code	Course Title	Hours/Week				L	T	P	C
		L	T	P	C				
UCD24V01T	Essentials of Artificial Intelligence	1	0	0	1				
UES24V01T	Environmental Studies	2	0	0	2				
UCD24V02T	Universal Human Values	2	0	0	2				
Total Learning Credits									5

Ability Enhancement Courses (A)									
Course Code	Course Title	Hours/Week				L	T	P	C
		L	T	P	C				
ULT24AE1J	Tamil - I								
ULH24AE1J	Hindi - I	2	0	2	3				
ULF24AE1J	French - I								
ULE24AE1J	English	2	0	2	3				
ULT24AE2J	Tamil - II								
ULH24AE2J	Hindi - II	2	0	2	3				
ULF24AE2J	French - II								
Total Learning Credits									9

Skill Enhancement Courses (S)									
Course Code	Course Title	Hours/Week				L	T	P	C
		L	T	P	C				
UCD24S01J	Verbal Ability and Skill Development	1	0	2	2				
UCD24S02L	Quantitative Aptitude and Logical Reasoning	0	0	2	1				
UCD24S03J	Industry Oriented Employability and Leadership Skills	1	0	2	2				
UEN24S01L	Communication Skills	0	0	4	2				
UCD24S04J	Career Readiness and Professional Skills	1	0	2	2				
Total Learning Credits									9

Ability Enhancement Courses (A)							Skill Enhancement Courses (S)						
Course Code	Course Title	Hours/Week			C		Course Code	Course Title	Hours/Week			C	
		L	T	P					L	T	P		
ULT24AE1J	Tamil - I						UCD24S01J	Verbal Ability and Skill Development	1	0	2	2	
ULH24AE1J	Hindi - I	2	0	2	3		UCD24S02L	Quantitative Aptitude and Logical Reasoning	0	0	2	1	
ULF24AE1J	French - I						UCD24S03J	Industry Oriented Employability and Leadership Skills	1	0	2	2	
ULE24AE1J	English	2	0	2	3		UEN24S01L	Communication Skills	0	0	4	2	
ULT24AE2J	Tamil - II						UCD24S04J	Career Readiness and Professional Skills	1	0	2	2	
ULH24AE2J	Hindi - II	2	0	2	3								
ULF24AE2J	French - II												
Total Learning Credits							Total Learning Credits						9

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Project Work / Internship (P)							Mandatory Courses (Y)						
Course Code	Course Title	Hours/Week			C		Course Code	Course Title	Hours/Week			C	
		L	T	P					L	T	P		
UCS24P01L	Internship - I	-	-	-	1		UNS24Y01L	NSS					
UCS24P02L	Internship - II	-	-	-	1		UNC24Y01L	NCC	0	0	0	0	
UCS24P03L	Project Work	0	0	4	2		UNO24Y01L	NSO					
UCS24P04L	Research Project and Dissertation						UYG24Y01L	YOGA					
UCS24P05L	Professional Internship	0	0	24	12		UMI24Y01L	My India Project	0	0	0	0	
Total Learning Credits							Total Learning Credits						0

Internship – I to be completed during Summer Vacation of the First year
Internship – II to be completed during Summer Vacation of the Second year

Students from other departments can enroll in Elective Courses (Minor) (E)
Students from all departments will be able to enroll in Multidisciplinary Courses (M).

8. Course Allocation across Semesters

Semester – I					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
ULT24AE1J	Tamil – I	2	0	2	3
ULH24AE1J	Hindi – I				
ULF24AE1J	French – I				
ULE24AE1J	English	2	0	2	3
UCS24101J	Digital Electronics	3	0	3	4
USA24102J	Programming for Problem Solving	3	0	3	4
UMS24101T	Discrete Mathematical Structures	4	0	0	4
UCD24S01J	Verbal Ability and Skill Development	1	0	2	2
Learning Credits					20

Semester – II					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
ULT24AE2J	Tamil – II	2	0	2	3
ULH24AE2J	Hindi – II				
ULF24AE2J	French – II				
USA24201J	Data Structures and Algorithms	3	0	3	4
UCS24202J	Object Oriented Programming using Java	3	0	3	4
UMS24202T	Mathematical Foundation	4	0	0	4
UCD24V01T	Essentials of Artificial Intelligence	1	0	0	1
	Multidisciplinary Course – I				3
UCD24S02L	Quantitative Aptitude and Logical Reasoning	0	0	2	1
UNS24Y01L	NSS	0	0	0	0
UNC24Y01L	NCC				
UNO24Y01L	NSO				
UYG24Y01L	YOGA				
Learning Credits					20

Total Learning Credits of Courses of FIRST Year = 40

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Semester – III					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
UCS24301J	Operating System	3	0	3	4
USA24302J	Database Management System	3	0	3	4
UMS24303T	Statistical Methods	4	0	0	4
	Minor Elective – I				4
UCS24P01L	Internship – I	0	0	0	1
	Multidisciplinary Course – II				3
Total Learning Credits					20

Semester – IV					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
USA24401J	Python Programming	3	0	3	4
UMS24D01T	Resource Management Techniques	4	0	0	4
UMS24D02T	Numerical Methods				
	Minor Elective – II				4
	Minor Elective – III				4
UEN24S01L	Communication Skills	0	0	4	2
UCD24S03J	Industry Oriented Employability and Leadership Skills	1	0	2	2
UMI24Y01L	My India Project	0	0	0	0
Total Learning Credits					20

Total Learning Credits of Courses of SECOND Year = 40

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Semester – V					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
UCS24D01J	Software Engineering	3	0	3	4
UCS24D02J	Computer Networks and Data Communication				
UCS24D03J	Object Oriented Analysis and Design	3	0	3	4
UCS24D04J	Machine Learning				
	Minor Elective – IV				4
	Multidisciplinary Course – III				3
UES24V01T	Environmental Studies	2	0	0	2
UCD24S04J	Career Readiness and Professional Skills	1	0	2	2
UCS24P02L	Internship – II	0	0	0	1
Total Learning Credits					20

Semester – VI					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
UCS24D05J	Fundamentals of Business Application Development	3	0	3	4
UCS24D06J	NLP and Language Models				
USA24D07J	Cloud Computing	3	0	3	4
UCS24D08J	Microprocessors and Embedded Systems				
	Minor Elective – V				4
	Minor Elective – VI				4
UCS24P03L	Project Work	0	0	4	2
UCD24V02T	Universal Human Values	2	0	0	2
Total Learning Credits					20

Total Learning Credits of Courses of THIRD Year = 40

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Semester – VII					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
UCS24D09J	Mobile Application Development	3	0	3	4
USA24D10T	Wireless Communication and Mobile Computing	4	0	0	
UCS24D11J	Web Application Framework	3	0	2	4
UCS24D12J	Computer Vision				
UCS24D13J	Big Data Analytics	3	0	2	4
UCS24D14J	Cryptography and Network Security				
	Minor Elective – VII				4
	Minor Elective – VIII				4
Learning Credits					20

Semester – VIII					
Course Code	Course Title	Hours/ Week			C
		L	T	P	
UCS24D15J	AI and Business Analytics	3	0	2	4
USA24D18J	Cyber Security				
USA24D20J	Blockchain Technology	3	0	2	4
UCS24D18T	Introduction to Cyber Physical Systems	4	0	0	
UCS24P04L	Research Project and Dissertation	0	0	24	12
UCS24P05L	Professional Internship				
Learning Credits					20

Total Learning Credits of Courses of FOURTH Year = 40

**8.1 Elective Courses (MINOR) offered by the Department of Computer Science to other Departments (Excluding Department of Computer Science)**

Elective Courses (Minor) (E)						
Semester	Course Code	Course Title	Sessions Per Week			C
			L	T	P	
III	UDS24E01J	Data Science and Analytics	3	0	3	4
IV	UDS24E02J	Data Analytics using Spreadsheet	3	0	3	4
IV	UDS24E03J	Essentials of Machine Learning	3	0	3	4
V	UDS24E04J	Programming using R	3	0	3	4
VI	UDS24E05J	Programming using Python	3	0	3	4
VI	UDS24E06J	Introduction to Deep Learning	3	0	3	4
VII	UDS24E07J	Applications of Computer Vision	3	0	3	4
VII	UDS24E08J	Data Visualization Tools	3	0	3	4
Total Learning Credits						32

8.2 Multidisciplinary Courses offered by the Department of Computer Science to all departments (Including Department of Computer Science)

Multidisciplinary Courses (M)						
Semester	Course Code	Course Title	Sessions Per Week			C
			L	T	P	
II	UCS24M01J	Introduction to ICT Tools	2	0	2	3
III	UCS24M02J	Fundamentals of Digital Marketing	3	0	1	3
V	UCS24M03J	Basics of Web Application Development	2	0	2	3
Total Learning Credits						9

